COMP3012 Assessment 1: Set Exercises

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a) ***Results table:***

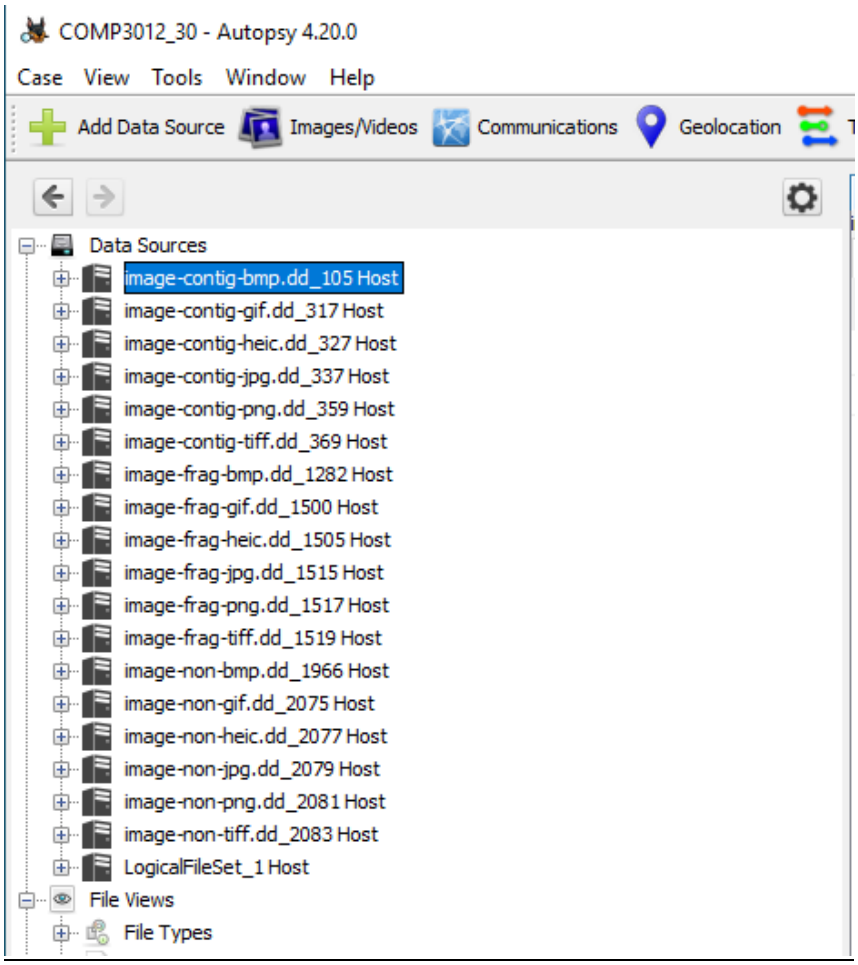
AUTOPSY			
Data Sets	***File Formats***	***Data Carved***	***Notes***
CONTINUOUS	BMP	6/6 Fully carved	All files clear and viewable
CONTINUOUS	GIF	6/6 Fully carved	All files clear and viewable
CONTINUOUS	HEIC	6/6 Partially carved	Unsupported - unable to open (assume all files clear)
CONTINUOUS	JPG	6/6 Fully carved	All files clear and viewable
CONTINUOUS	PNG	6/6 Fully carved	All files clear and viewable
CONTINUOUS	TIFF	6/6 Fully carved	All files clear and viewable
NON-ALIGNED	BMP	0/6 None carved	4 .txt files present
NON-ALIGNED	GIF	0/6 None carved	No files present
NON-ALIGNED	HEIC	0/6 None carved	No files present
NON-ALIGNED	JPG	0/6 None carved	No files present
NON-ALIGNED	PNG	0/6 None carved	No files present
NON-ALIGNED	TIFF	0/6 None carved	6 .txt files, 2 .h files, 2 .mp3 files
FRAGMENTED	BMP	6/6 Partially carved	Present BMP files are partially corrupted, 9 .txt files also present
FRAGMENTED	GIF	1/6 Partially carved, 5/6 Not carved	Present GIF file is partially corrupted
FRAGMENTED	HEIC	6/6 Partially carved	Unsupported - unable to open (assume corrupted)
FRAGMENTED	JPG	0/6 None carved	No files present
FRAGMENTED	PNG	0/6 None carved	No files present
FRAGMENTED	TIFF	0/6 None carved	No files present

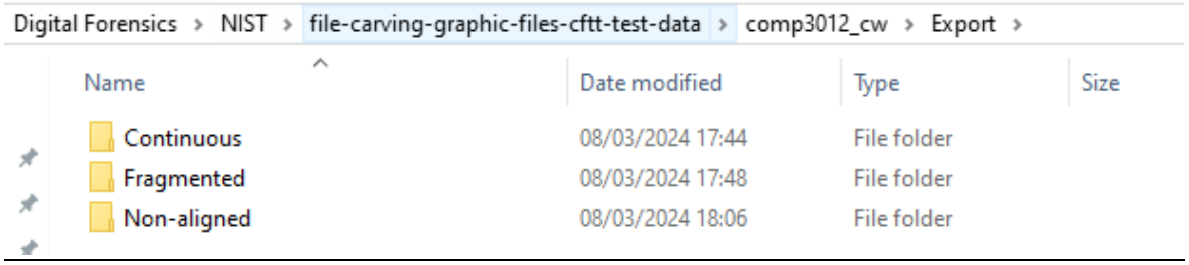
SCALPEL			
Data Sets	***File Formats***	***Data Carved***	***Notes***
CONTINUOUS	BMP	1/6 Partially carved, 5/6 Not carved	Unable to open file present
CONTINUOUS	GIF	3/6 Fully carved, 2/6 Partially carved, 1/6 Not carved	Partially carved files are partially corrupted
CONTINUOUS	HEIC	0/6 None carved	No files present
CONTINUOUS	JPG	2/6 Fully carved, 4/6 Partially carved	Partially carved files are fully corrupted - unable to open
CONTINUOUS	PNG	6/6 Partially carved	Unable to open all files present
CONTINUOUS	TIFF	6/6 Fully carved	All files clear and viewable
NON-ALIGNED	BMP	1/6 Partially carved, 5/6 Not carved	File is fully corrupted - unable to open
NON-ALIGNED	GIF	3/6 Fully carved, 2/6 Partially carved, 1/6 Not carved	Partially carved files are partially corrupted
NON-ALIGNED	HEIC	0/6 None carved	No files present
NON-ALIGNED	JPG	2/6 Fully carved, 4/6 Partially carved	Partially carved files are fully corrupted - unable to open
NON-ALIGNED	PNG	6/6 Partially carved	Unable to open all files present
NON-ALIGNED	TIFF	6/6 Fully carved	All files clear and viewable
FRAGMENTED	BMP	1/6 Partially carved, 5/6 Not carved	Unable to open file present
FRAGMENTED	GIF	5/6 Partially carved, 1/6 Not carved	All files present are partially corrupted
FRAGMENTED	HEIC	0/6 None carved	No files present
FRAGMENTED	JPG	1/6 Fully carved, 1/6 Partially carved, 4/6 Not carved	1/6 file partially corrupted, 4/6 files fully corrupted - unable to open
FRAGMENTED	PNG	6/6 Partially carved	All files unable to open
FRAGMENTED	TIFF	6/6 Partially carved	All files unable to open

FTK			
Data Sets	***File Formats***	***Data Carved***	***Notes***
CONTINUOUS	BMP	0/6 None carved	No files present
CONTINUOUS	GIF	1/6 Fully carved, 5/6 Partially carved	2/6 partially carved files are partially corrupted and 3/6 are fully corrupted
CONTINUOUS	HEIC	0/6 None carved	No files present
CONTINUOUS	JPG	6/6 Fully carved	3 additional partially carved duplicates (lower quality) exist along with fully carved files
CONTINUOUS	PNG	6/6 Fully carved	All files clear and viewable
CONTINUOUS	TIFF	1/6 Fully carved, 5/6 Not carved	File present is clear and viewable
NON-ALIGNED	BMP	0/6 None carved	No files present
NON-ALIGNED	GIF	1/6 Fully carved, 5/6 Partially carved	2/6 partially carved files are partially corrupted and 3/6 are fully corrupted
NON-ALIGNED	HEIC	0/6 None carved	No files present
NON-ALIGNED	JPG	6/6 Fully carved	3 additional partially carved duplicates (lower quality) exist along with fully carved files
NON-ALIGNED	PNG	6/6 Fully carved	All files clear and viewable
NON-ALIGNED	TIFF	1/6 Fully carved, 5/6 Not carved	File present is clear and viewable
FRAGMENTED	BMP	0/6 None carved	No files present
FRAGMENTED	GIF	6/6 Partially carved	3/6 are partially corrupted, 3/6 are fully corrupted - unable to open
FRAGMENTED	HEIC	0/6 None carved	No files present
FRAGMENTED	JPG	2/6 Fully carved, 4/6 Partially carved	3 additional partially carved duplicates exist, partially carved files are partially corrupted
FRAGMENTED	PNG	6/6 Partially carved	All files are partially corrupted
FRAGMENTED	TIFF	0/6 None carved	No files present

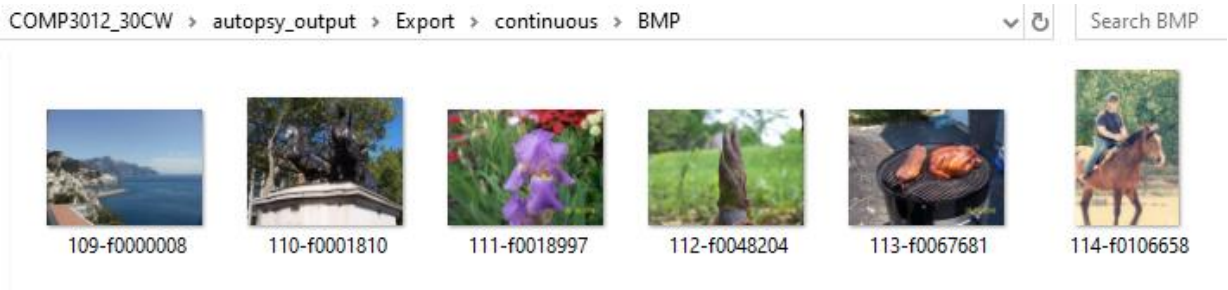
b) ***Analysis:***

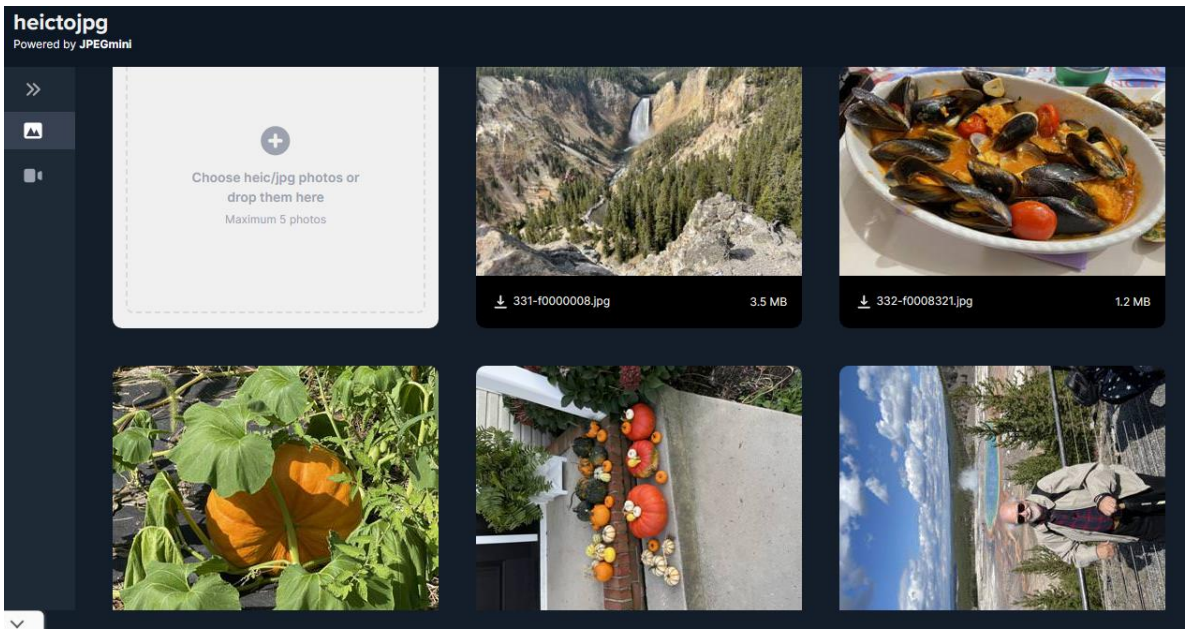
Autopsy



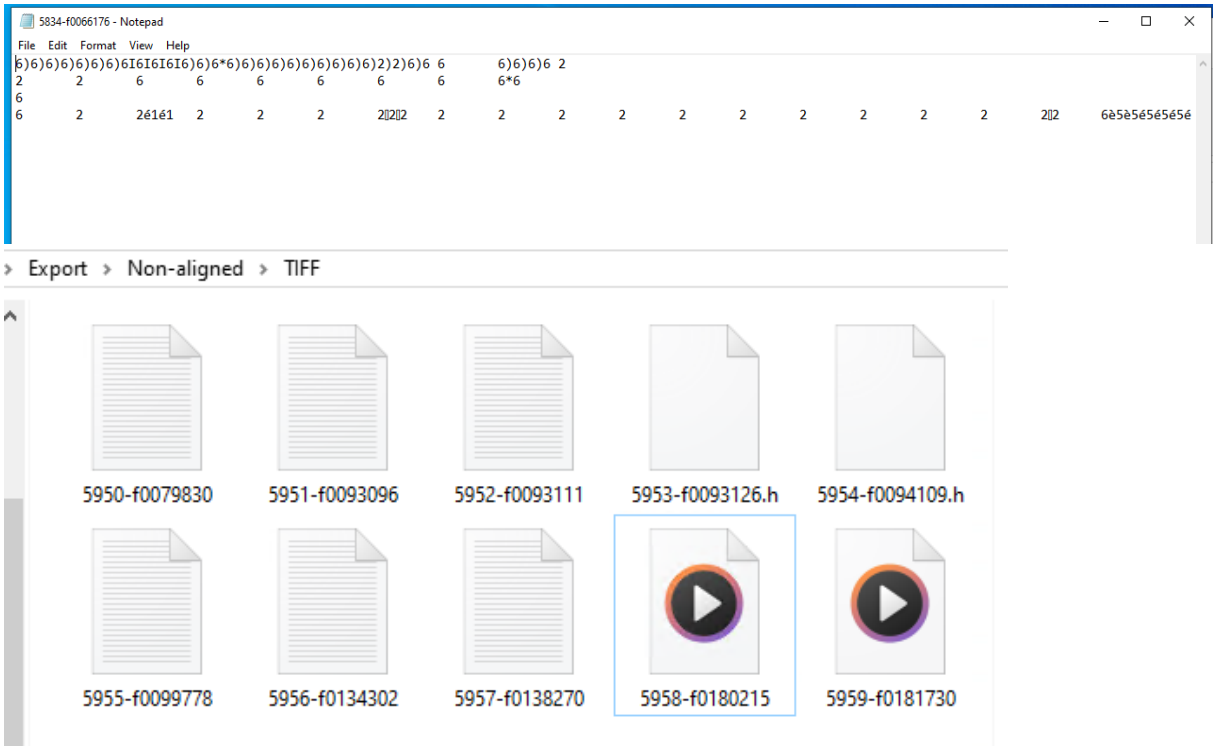


Autopsy was very successful in data carving continuous files. Output had images identical to source files. HEIC was unviewable due to being unsupported, so I exported them to an external HEIC viewer/converter. Results showed images identical to the source images. As these images were stored in a continuous, unfragmented manner, they were easily identified and reconstructed by the tool.

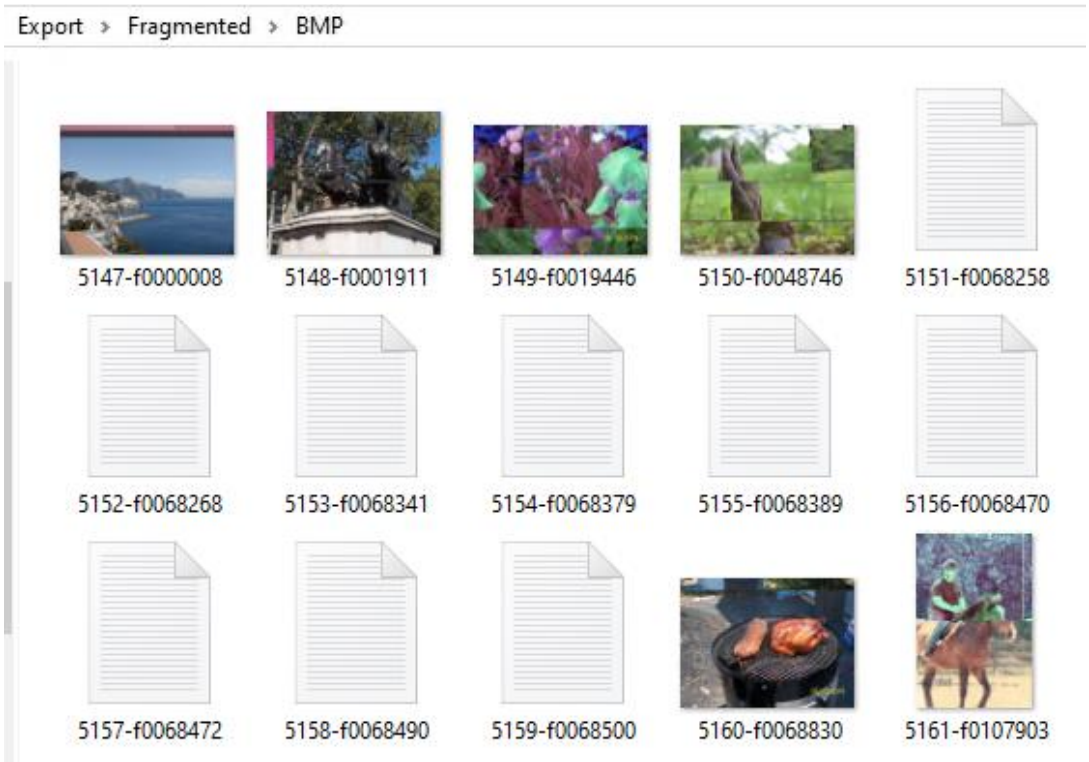




It wasn't too successful for non-aligned as 4/6 file types returned nothing. Only BMP and TIFF returned outputs, containing non-image file types - text, header and mp3 files. The tool could have extracted embedded text within the image, or due to metadata extraction (timestamps or settings of the device's camera settings, etc). Autopsy relies on recognizing file patterns, so if the files don't have headers/footers intact in their contiguous blocks, it's harder to identify them.



It was more successful for fragmented compared to non-aligned. Only BMP, GIF and HEIC were partially carved. Image files present in BMP and GIF were partially corrupt as they were not identical to the source images - contained blank spaces, pixelation and colour changes. BMP produced additional text files. Attempted to export HEIC images to the viewer used previously but was unsuccessful as I assume the software isn't that advanced. No evidence, but I would like to assume these would possibly also be partially corrupt image files. Even if autopsy's speciality is to identify and reconstruct fragmented data, some image files in the dataset may be extremely fragmented, so more difficult to recover.



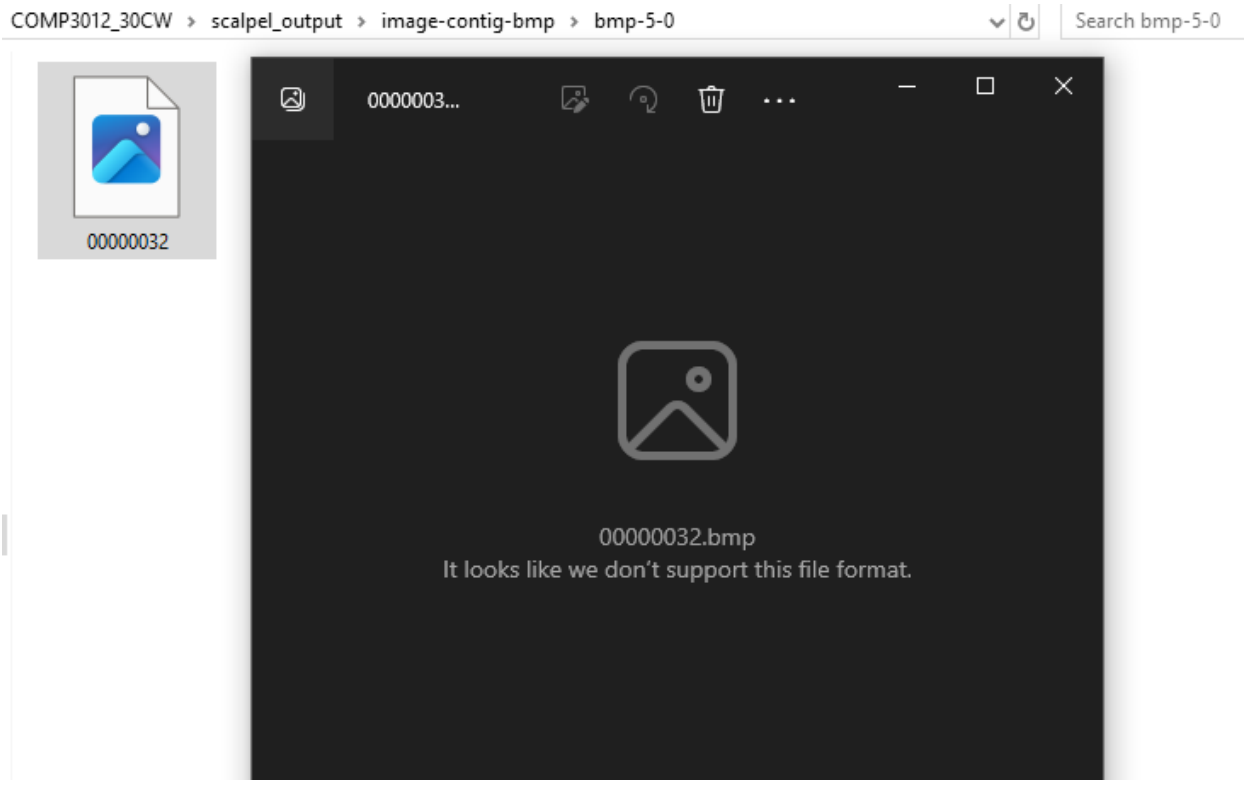
Scalpel

```
C:\Users\student\Desktop\Digital Forensics\scalpel-2.0>scalpel image-contig-bmp.dd
Scalpel version 2.0
Written by Golden G. Richard III and Lodovico Marziale.
Multi-core CPU threading model enabled.
Initializing thread group data structures.
Creating threads...
Thread creation completed.

Opening target "C:\Users\student\Desktop\Digital Forensics\scalpel-2.0\image-contig-bmp.dd"

Image file pass 1/2.
image-contig-bmp.dd:  13.0% |********                                        |   10.0 MB    00:00 ETAimage-contig-bmp.dd:  26.1% |*****************
    |   20.0 MB    00:00 ETAimage-contig-bmp.dd:  39.1% |***************************                                        |   30.0 MB    00:00 ETAimage-contig-bmp.dd:  52.2% |**********************************
                                |   40.0 MB    00:00 ETAimage-contig-bmp.dd:  65.2% |*****************************************                                        |   50.0 MB    00:00 ETAimage-contig-bmp.dd:  78.3% |*******
**********************************************                                        |   60.0 MB    00:00 ETAimage-contig-bmp.dd:  91.3% |*****************************************************************                                        |   70.0 MB    00:00 ETAimage-co
ntig-bmp.dd: 100.0% |*****************************************************************|   76.6 MB    00:00 ETAAllocating work queues...
Work queues allocation complete. Building work queues...
Work queues built.  Workload:
gif with header "\x47\x49\x46\x38\x37\x61" and footer "\x00\x3b" --> 0 files
gif with header "\x47\x49\x46\x38\x39\x61" and footer "\x00\x00\x3b" --> 0 files
jpg with header "\xff\xd8\xff\xe0\x00\x10" and footer "\xff\xd9" --> 0 files
jpg with header "\xff\xd8\xff\xe1" and footer "\xff\xd9" --> 0 files
png with header "\x50\x4e\x47?" and footer "\xff\xfc\xfd\xfe" --> 32 files
bmp with header "BM?}\x00\x00\x00" and footer "" --> 1 files
tif with header "\x49\x49\x2a\x00" and footer "" --> 0 files
tif with header "\x4D\x4D\x00\x2A" and footer "" --> 2 files
Carving files from image.
Image file pass 2/2.
image-contig-bmp.dd:  13.0% |********                                        |   10.0 MB    00:00 ETAimage-contig-bmp.dd:  52.2% |**********************************
    |   40.0 MB    00:00 ETAimage-contig-bmp.dd:  65.2% |*****************************************                                        |   50.0 MB    00:00 ETAimage-contig-bmp.dd:  78.3% |**********************************
*****************                                        |   60.0 MB    00:00 ETAimage-contig-bmp.dd:  91.3% |*****************************************************************                                        |   70.0 MB    00:00 ETAimage-contig-bmp.dd: 100.0% |*******
*****************************************************|   76.6 MB    00:00 ETAProcessing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 35, elapsed  = 1 secs.
```

› scalpel_output ›

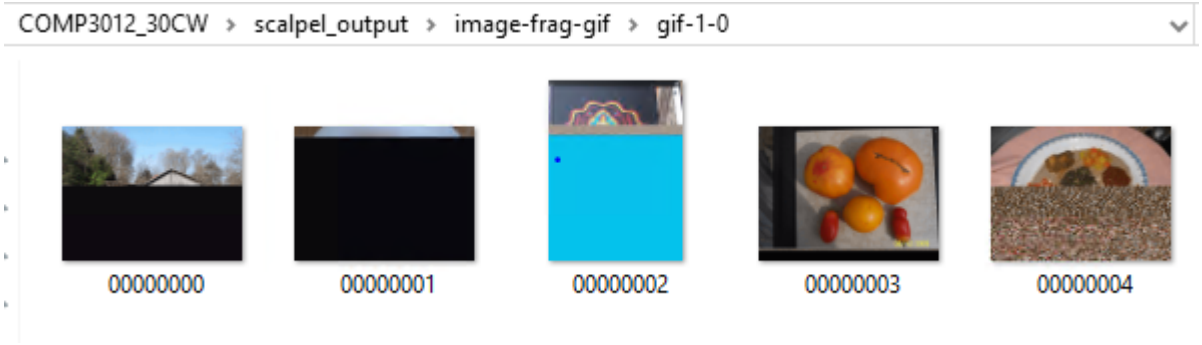
Name	Date modified	Type	Size
image-contig-bmp	13/03/2024 19:02	File folder	
image-contig-gif	13/03/2024 19:02	File folder	
image-contig-heic	13/03/2024 19:02	File folder	
image-contig-jpg	13/03/2024 19:02	File folder	
image-contig-png	13/03/2024 19:02	File folder	
image-contig-tiff	13/03/2024 19:02	File folder	
image-frag-bmp	13/03/2024 19:02	File folder	
image-frag-gif	13/03/2024 19:02	File folder	
image-frag-heic	13/03/2024 19:02	File folder	
image-frag-jpg	13/03/2024 19:02	File folder	
image-frag-png	13/03/2024 19:02	File folder	
image-frag-tiff	13/03/2024 19:02	File folder	
image-non-bmp	13/03/2024 19:02	File folder	
image-non-gif	13/03/2024 19:02	File folder	
image-non-heic	13/03/2024 19:02	File folder	
image-non-jpg	13/03/2024 19:02	File folder	
image-non-png	13/03/2024 19:02	File folder	
image-non-tiff	13/03/2024 19:02	File folder	
scalpel_log	13/03/2024 19:02	Text Document	56 KB

After customising the scalpel.conf file and running on terminal, outputs indicate scalpel is moderately good at data carving continuous files. The only fully carved file type was TIFF. Some partially carved files were fully corrupt – unable to open, so may need to further alter the configuration file. HEIC had no outputs. Scalpel's carving algorithm identifies file headers, footers and internal structures within continuous files and extracts in a systematic manner, allowing reconstruction. This works well with continuous data as they are stored in contiguous blocks.

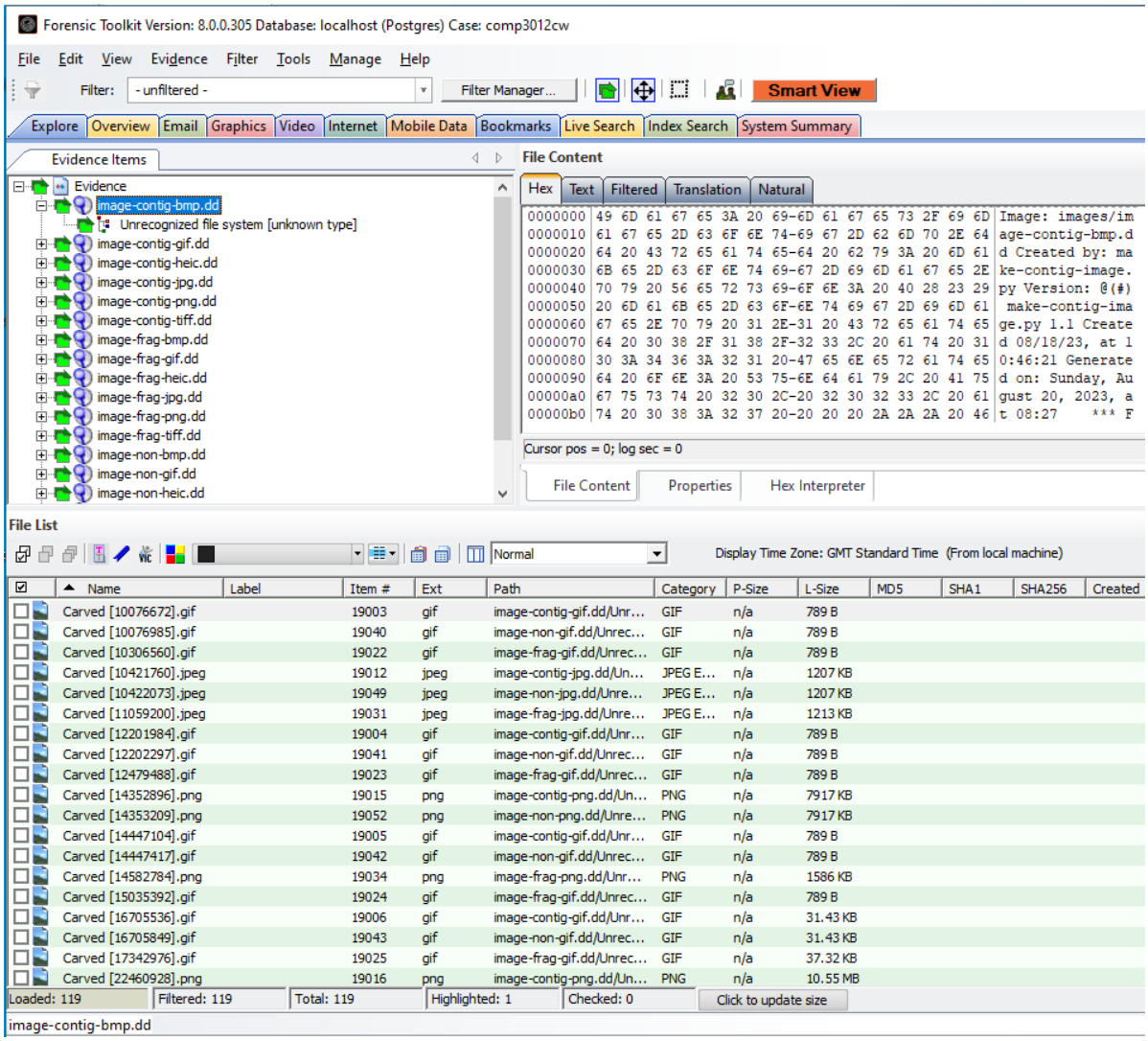


Performed similarly for non-aligned files, as again, the only file type fully carved was TIFF, HEIC had no output, and other types were partially carved. With non-aligned files, the tool's ability to examine file system metadata allows it to identify and recover things such as file location and characteristics (e.g. byte patterns) from the file allocation tables, directory entries and other metadata provided.

Scalpel performed quite poorly with carving fragmented files as only 1/6 of JPG was fully carved. It operates by sequentially reading data, so may have struggled with accurately identifying and reconstructing the heavily fragmented files as fragments may be scattered across multiple non-contiguous locations across the disk, or if metadata is incomplete or corrupted. It can only handle fragmentation to a certain extent, so it's harder to get precise outputs without further customising the configuration file.



FTK



FTK worked well for data carving continuous files. It fully carved JPG and PNG and 1/6 TIFF, partially for GIF, but not BMP or HEIC. Worked well as FTK analyses header and footer position, along with file signature matching and sequential data reading through the entire dataset. FTK also produced thumbnails (described as lower quality duplicates in table), which could be due to a number of reasons. For example, some images contain embedded thumbnails which are extracted by FTK as separate files, or could contain metadata about thumbnails associated with the files that are extracted separately, etc.

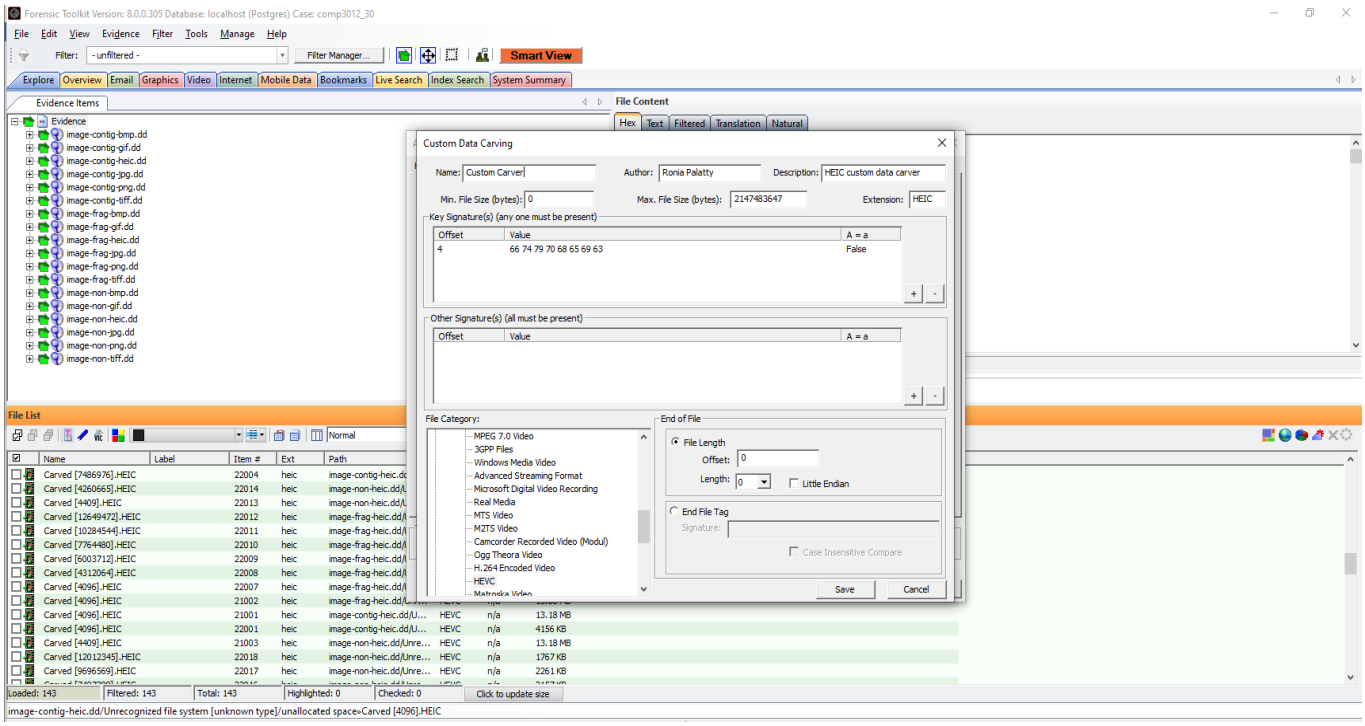
It worked similarly for non-aligned files. Features such as its file signature analysis, sequential data reading and data structure analysis allows to identify the non-aligned files' location and characteristics, helping to ensure the files are recovered.

FTK wasn't too successful with fragmented files. It only managed to fully carve 2/6 JPG files. BMP, HEIC and TIFF had no output. Similar with the other tools, it could be because of the complexity of the fragmented files or if the file system metadata is incomplete or corrupted. Especially as the data will not be in contiguous blocks, it is harder to recover and may need the help of other FTK tools alongside the carver.

Conclusion:

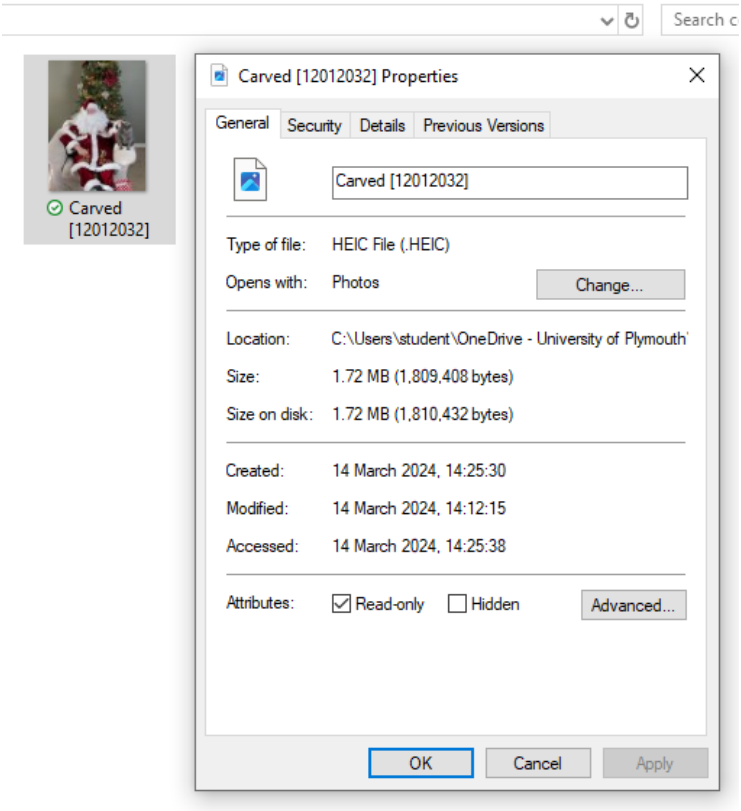
Each carving tool has their strengths and weaknesses. It also depends on other aspects such as the files being used (e.g. the intensity of fragmentation), the version being used, etc. For example, Autopsy did well for carving continuous files, but badly for the other datasets. Scalpel and FTK are both good tools in the sense that you can customise the file signatures, which will help the tool recognise the file patterns better, allowing efficient reconstruction. If I had to choose it'd be FTK as I am taking other aspects into consideration such as user interaction and viewability into consideration. Scalpel was a bit tedious to enter and view outputs and would require an external script to run it all at once, whereas FTK does that automatically. FTK also has more analysis tools that can help further with image analysis.

c) ***Custom Carver:***



```
[4 byte offset]                    [4 byte offset]
66 74 79 70 68 65 69 63            ftypheic
```

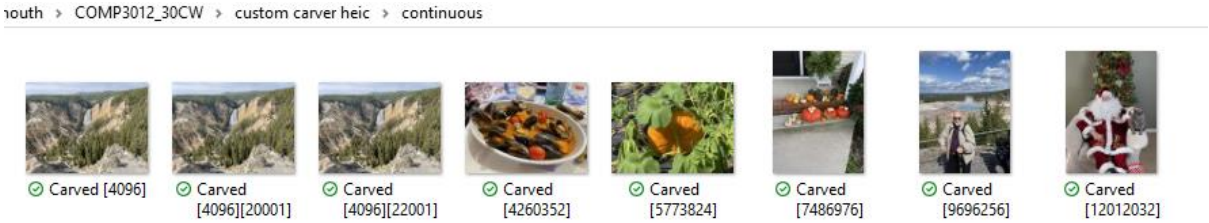
HEIC High Efficiency Image Container (HEIC), holding one or more High Efficiency Image File (HEIF) — aka H.265 — images, created using the High Efficiency Video Coding (HEVC) algorithm. Developed by the Moving Pictures Experts Group (MPEG) and adopted by Apple as the standard image file format in 2017.



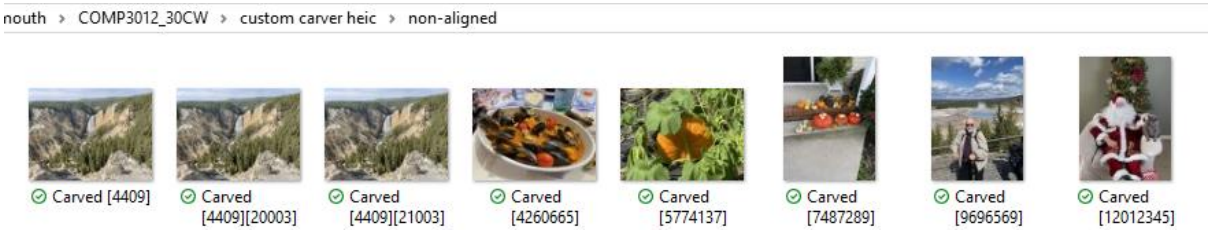
For this task, I created a custom data carver in FTK. I debated on using scalpel but prefer FTK's interface as it makes viewing easier. Starting by understanding what exactly HEIC files are and why they didn't work well with the other data carving tools used, I understood that HEIC files are usually heavily fragmented, which makes it difficult for the tools to correctly identify and precisely reconstruct the HEIC files.

Then I imported all the files into FTK and selected the data carving option in the additional analysis miscellaneous section. There I selected the option to create my own custom carver and I entered the details required such as the offset number and file signature values that will uniquely identify HEIC files. After, I confirmed everything and ran analysis with the custom data carver selected.

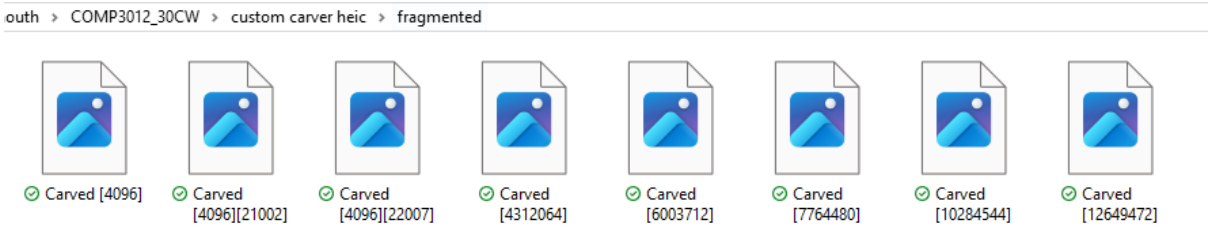
The custom carver successfully carved the continuous dataset and produced images identical to the source images provided. It also produced thumbnails which could be a result of embedded thumbnails in the files or thumbnail data in the metadata that was extracted separately. As they are stored in a continuous block in the storage medium without fragmentation, it is easier for the carver to identify and reconstruct files.



I got the same results for non-aligned as I did with the continuous dataset. It also produced thumbnails like the continuous dataset. All images matched the source files. This is because even if non-aligned is in non-contiguous blocks, it may still have intact file headers and footers and show patterns, which allows the carver to identify and reconstruct it without relying too much on the metadata.



The custom carver did poorly with the fragmented HEIC files as all of them were corrupted and unable to open. This may be because with fragmented, all data segments (including header and footer) are in a random disordered scattered manner, making it harder for the carver to identify and extract. There is a possibility it could be partially carved, but I don't have access to any viewers advanced enough to open partially corrupt HEIC files.



Website used to find file signature:

https://www.garykessler.net/library/file_sigs.html